

Comparison of Network Topologies

Star, Bus, Ring, and Mesh Topologies — Pros, Cons, Best Use Cases, and App/Service Examples

Network Topology	Pros	Cons	Best Use Case	Example from Apps/Services
Star Topology	1. Easy to install and manage. 2. Failure of one end device usually does not affect others. 3. Easy to add or remove devices. 4. Fault detection is simple. 5. Good performance with a modern switch.	1. Central switch/hub is a critical point of failure. 2. Requires more cable than bus topology. 3. Installation cost can be higher. 4. Network size depends on central device capacity. 5. Switch failure may stop the whole network.	Office, school, and computer-lab LANs where many devices connect to a central switch.	Instagram — simplified analogy: users access centrally managed service infrastructure to upload posts, view feeds, and send requests.
Bus Topology	1. Simple network design. 2. Uses less cable than star topology. 3. Low installation cost for small networks. 4. Easy to understand and set up in limited environments. 5. Does not require a central switch in a pure bus network.	1. Backbone cable failure can stop the whole network. 2. Difficult to troubleshoot faults. 3. Performance decreases as traffic increases. 4. Traditional shared media can experience collisions. 5. Limited cable length and number of devices.	Small temporary or legacy networks with few devices and low traffic.	IPL live streaming — simplified analogy: one live content source is distributed toward many viewers, similar to shared or broadcast-style communication.
Ring Topology	1. Data can move in an orderly sequence. 2. Predictable performance under controlled traffic. 3. Reduced collision risk in token-based rings. 4. Each node can help pass data to the next node. 5. Can provide fair transmission access.	1. A single node or link failure may disrupt a basic single ring. 2. Adding or removing devices can interrupt the network. 3. Troubleshooting can be difficult. 4. Delay increases as more nodes are added. 5. Less common in modern general-purpose LANs.	Industrial or specialized networks where predictable sequential communication is useful.	Zomato — simplified analogy: an order can move through stages such as customer → restaurant → preparation → delivery partner → customer.
Mesh Topology	1. Very high reliability. 2. Multiple paths are available for data. 3. Excellent fault tolerance. 4. One link failure usually does not stop the entire network. 5. Suitable for critical systems requiring redundancy.	1. Expensive to install. 2. Requires many links or connections. 3. Configuration is complex. 4. Maintenance can be difficult. 5. Full mesh becomes increasingly complex as nodes increase.	Critical, distributed, or high-availability networks where service continuity is important.	Paytm — simplified analogy: digital payment services require resilient infrastructure and alternative paths or components so transactions can remain available during failures.